# The Air We Breathe

Pupil sheet

**Name:** _____ **Date:** _____

**Safety:** Breathe only into your own bottle, not onto the box; the teacher opens the window.

## 1. What I see on the screen

On the screen you see the **$CO_2$** card (carbon dioxide, in ppm — "parts per million") and its chart underneath. You start with **Start**; you move on with the green button at the bottom right, and the last button, a blue one, says **Finish**. You also type your prediction into the app before measuring. At each measurement you press **Capture**.

**The sensor is slow in a particular way:** it gives a new value once every **five seconds**, so the line climbs in steps. It is not broken.

**The most important rule:** when we measure the room's air, we **stay a step back** and do not lean over the box. Otherwise we measure our own air, not the room's.

## 2. What I do on my own

**First I predict. I do not change my prediction after I measure.**

**1.** If I breathe at the sensor, the number will: ☐ ↑ ☐ ↓ ☐ →
**2.** Once I stop, it will come back in: ☐ a second ☐ about ten seconds ☐ one to three minutes

because _____

_____

**3.** Put an arrow on the line where you think our room is today:

| 420 | 800 | 1200 | 2000 ppm |
|---|---|---|---|
| outdoors Excellent | Good | Moderate | Poor |

**While I wait for the recovery, I draw the shape of the line.** How it went up, how it comes down.

**Now I measure.** If the number is too big for me to write, I just circle the word in the last column.

| Moment | $CO_2$ (ppm) | Circle one |
|---|---|---|
| The room's air, at the start | | Excellent / Good / Moderate / Poor |
| Just after I breathed | | Excellent / Good / Moderate / Poor |
| After it came back | | Excellent / Good / Moderate / Poor |
| After we opened the window | | Excellent / Good / Moderate / Poor |
| A second breath, the same | | close to the first? yes / no |

**Working it out.** The biggest: _____ ppm the smallest: _____ ppm
**How long the recovery took:** _____ minutes and _____ seconds. Children in the room: _____

## 3. The conclusion I write

I predicted that _____

_____

The data show that _____

_____

It went up fast and came down slowly because _____

_____

**Our class rule:** when, and who, opens the window? Use your numbers.

_____

**Exit ticket**

Write one thing that surprised you today.

_____